MATHEMATICS II FOR DATA SCIENCE
GA-3 SOLUTIONS MAY 2024

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1. Question

Which of the following sets with the given addition and scalar multiplication operations (scalars are real numbers in every case) do not form vector spaces?

- [ ] $V_1 = \{(x, y) | x, y \in \mathbb{R}\}$
  $Addition: (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, 1);\ (x_1, y_1), (x_2, y_2) \in V_1$
  $Scalar\ multiplication: c(x, y) = (cx, 1);\ (x, y) \in V_1,\ c \in \mathbb{R}$

- [ ] $V_2 = \{(x, y) | x, y \in \mathbb{R}\}$
  $Addition: (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2);\ (x_1, y_1), (x_2, y_2) \in V_2$
  $Scalar\ multiplication: c(x, y) = (cx, 0);\ (x, y) \in V_2,\ c \in \mathbb{R}$

- [ ] $V_3 = \{(x, y) | x, y \in \mathbb{R}\}$
  $Addition: (x_1, y_1) + (x_2, y_2) = (x_1 + x_2 + y_1 + y_2, x_1 + x_2 + y_1 + y_2);$
  $(x_1, y_1), (x_2, y_2) \in V_3$
  $Scalar\ multiplication: c(x, y) = (cx, cy);\ (x, y) \in V_3,\ c \in \mathbb{R}$

- [ ] $V_4 = \{(x, y, z) | x, y, z \in \mathbb{R}, x + y = z\}$
  $Addition: (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2);$
  $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_4$
  $Scalar\ multiplication: c(x, y, z) = (cx, cy, cz);\ (x, y, z) \in V_4,\ c \in \mathbb{R}$

1.1. Answers:

- $V_1 = \{(x, y)|x, y \in \mathbb{R}\}$
  $Addition: (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, 1);\ (x_1, y_1), (x_2, y_2) \in V_1$
  $Scalar\ multiplication: c(x, y) = (cx, 1);\ (x, y) \in V_1,\ c \in \mathbb{R}$

- $V_2 = \{(x, y)|x, y \in \mathbb{R}\}$
  $Addition: (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2);\ (x_1, y_1), (x_2, y_2) \in V_2$
  $Scalar\ multiplication: c(x, y) = (cx, 0);\ (x, y) \in V_2,\ c \in \mathbb{R}$

- $V_3 = \{(x, y)|x, y \in \mathbb{R}\}$
  $Addition: (x_1, y_1) + (x_2, y_2) = (x_1 + x_2 + y_1 + y_2, x_1 + x_2 + y_1 + y_2);$
  $(x_1, y_1), (x_2, y_2) \in V_3$
  $Scalar\ multiplication: c(x, y) = (cx, cy);\ (x, y) \in V_3,\ c \in \mathbb{R}$

1.2. Solution

- For each option, we apply addition and scalar multiplication as defined for that option.
- A non-empty set of vectors is not a vector space if it fails to satisfy even one of the nine vector space axioms.
- There is no zero vector 0_v in V_1 for which $u + 0_v = u$, for all $u \in V_1$. Let us take $u = (3, 3)$ and $0_v = (p, q)$, where $p, q \in \mathbb{R}$.

$$u + 0_v = (3, 3) + (p, q) = (3 + p, 1)$$

  We can see that $u + 0_v = u$ is not possible for any $(p, q) \in V_1$. The second coordinate of $u + 0_v$ is 1. But the second coordinate of u is 3. Since $3 \neq 1$, V_1 fails to have a zero vector 0_v which satisfies $u + 0_v = u$. V_1 is not a vector space. Hence, **option 1** is correct.
- V_2 fails to satisfy the property of <u>unity</u>. V_2 fails to satisfy $1 \cdot u = u$, where $u \in V_2$. Let us take $u = (2, 3)$.

$$1 \cdot u = 1 \cdot (2, 3) = (2, 0) \neq u$$

  V_2 is not a vector space. Hence, **option 2** is correct.
- V_3 does not satisfy the property of <u>distributivity</u> : $(a + b) \cdot u = au + bu$ for all $u \in V_3$ and for all $a, b \in \mathbb{R}$. Let us take $u = (1, 2)$. $a = 3$ and $b = 2$.

$$(a + b) \cdot u \Longrightarrow (3 + 2) \cdot (1, 2) = 5 \cdot (1, 2) = (5, 10) \qquad \text{(LHS)}$$
$$au + bu \Longrightarrow 3 \cdot (1, 2) + 2 \cdot (1, 2) = (3, 6) + (2, 4) = (15, 15) \qquad \text{(RHS)}$$

  $LHS \neq RHS$. Hence, **option 3** is correct.
- V_4 satisfies all the vector space axioms. V_4 is a vector space. Let us take $u = (x, y, z)$, $v = (p, q, r)$ and $w = (l, m, n)$ such that $u, v, w \in V_4$. Let $a, b \in \mathbb{R}$

VECTOR SPACE AXIOMS	PROOF
$u + v \in V_4$	$(x, y, z) + (p, q, r) = (x + p, y + q, z + r)$ $x + p \in \mathbb{R},\ y + q \in \mathbb{R},\ z + r \in \mathbb{R}$. Therefore, $u + v \in V_4$

$a \cdot u \in V_4$	$a.(x, y, z) = (ax, ay, az)$ $ax \in \mathbb{R},\ ay \in \mathbb{R},\ az \in \mathbb{R}$. Therefore, $a \cdot y \in V_4$
$u + v = v + u$	$LHS \Longrightarrow (x, y, z) + (p, q, r) = (x + p, y + q, z + r)$ $RHS \Longrightarrow (p, q, r) + (x, y, z) = (p + x, q + y, r + z)$ $LHS = RHS$
Existence of zero vector	Let $0_v = (i, j, k)$. $u + 0_v \Longrightarrow (x, y, z) + (i, j, k) = (x + i, y + j, z + k)$ Equate $u + 0_v$ with u. • $(x + i, y + j, z + k) = (x, y, z)$ • $i = 0, j = 0, k = 0.$ • $0_v = (0, 0, 0)$ • $0_v \in V_4$
$1 \cdot u = i$	$1 \cdot u = 1 \cdot (x, y, z) = (x, y, z) = u$
$(u + v) + w = y + (v + w)$	$(u + v) + w = ((x, y, z) + (p, q, r)) + (l, m, n)$ $(u + v) + w = (x + p, y + q, z + r) + (l, m, n)$ $(u + v) + w = (x + p + l, y + q + m, z + r + n)$ (1) $u + (v + w) = (x, y, z) + ((p, q, r) + (l, m, n))$ $u + (v + w) = (x, y, z) + (p + l, q + m, r + n)$ $u + (v + w) = (x + p + l. y + q + m, z + r + n)$ (2) From (1) and (2) we can conclude that : – $(u + v) + w = y + (v + w)$
$(ab) \cdot u = a(b \cdot u)$	$LHS \Longrightarrow (ab) \cdot (x, y, z) = (abx, aby, abz)$ $RHS \Longrightarrow a(b \cdot (x, y, z)) = a \cdot (bx, by, bz) = (abx, aby, abz)$ $LHS = RHS$
Existence of additive inverse	Let $u' = (d, e, f)$. $u + u' = (x, y, z) + (d, e, f) = (x + d, y + e, z + f)$ Equate $u + u'$ with 0_v • $(x + d, y + e, z + f) = (0, 0, 0)$ • $d = -x, e = -y, f = -z$ • $u' = (-x, -y, z)$ • $u' \in V_4$

$(a+b)\cdot u = au + bu$	$(a+b)\cdot u = (a+b)\cdot(x,y,z)$ $(a+b)\cdot u = (ax+bx, ay+by, az+bz)$ (1) $au+bu = a\cdot(x,y,z) + b\cdot(x,y,z)$ $au+bu = (ax,ay,az)+(bx,by,bz)$ $au+bu = (ax+bx, ay+by, az = bz)$ (2) From (1) and (2), we can conclude that :– $(a+b)\cdot u = au+bu$
$a\cdot(u+v) = au+av$	$a\cdot(u+v) = a\cdot((x,y,z)+(p,q,r))$ $a\cdot(u+v) = a\cdot(x+p, y+q, z+r)$ $a\cdot(u+v) = (ax+ap.ay+aq+az+ar)$ (1) $au+av = a\cdot(x,y,z)+a\cdot(p,q,r)$ $au+av = (ax,ay,az)+(ap,aq,ar)$ $au+av = (ax+ap, ay+aq, az+ar)$ (2) From (1) and (2), we can conclude that :– $a\cdot(u+v) = au+av$

Hence, **option 4** is incorrect.

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2. Question

Choose the set of correct options

- ☐ If V is a real vector space, then $(\alpha+\beta)(x+y) = \alpha x + \beta y + \alpha y + \beta x$, for all $\alpha, \beta \in \mathbb{R}$ and $x, y \in V$.
- ☐ A vector space can have more than one zero vector.
- ☐ (-1, 0 ,0), (-1, 1, -1) and (0, 2, 3) are linearly independent vectors in $\mathbb{R}^3$.
- ☐ $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ are linearly dependent vectors in $M_{2\times 2}(\mathbb{R})$.

2.1. Answers :

- If V is a real vector space, then $(\alpha+\beta)(x+y) = \alpha x + \beta y + \alpha y + \beta x$, for all $\alpha, \beta \in \mathbb{R}$ and $x, y \in V$.
- (-1, 0 ,0), (-1, 1, -1) and (0, 2, 3) are linearly independent vectors in $\mathbb{R}^3$.

2.2. Solution

- **Option 1** is correct. $(\alpha+\beta)+(x+y) = \alpha(x+y)+\beta(x+y) = \alpha x + \alpha y + \beta x + \beta y$
- **Option 2** is incorrect. A vector space can have only one zero vector because the zero vector of a

vector space is unique.

- **Option 3** is correct. Let matrix A have all the given vectors in its columns.

$$A = \begin{bmatrix} -1 & -1 & 0 \\ 0 & 1 & 2 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\begin{aligned} det(A) &= -1((1 \times 3) - (-1 \times 2)) \\ det(A) &= -1(3 + 2) \\ det(A) &= -5 \end{aligned}$$

Since $det(A) \neq 0$, we can say that the given vectors are linearly independent.

- **Option 4** is incorrect.

$$a\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} + b\begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} + c\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -a - c & -2b + c \\ b + c & a - c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

We get the following equations : –

  - $-a - c = 0$
  - $-2b + c = 0$
  - $b + c = 0$
  - $a - c = 0$

Let us solve the above system of linear equations by Gaussian Elimination method.

$$A = \begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}, \; x = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \; b = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

To solve $Ax = b$, we need not augment A and b. There is no non-zero element in b. Elementary row operations would not affect the column which corresponds to b.

Apply elementary row operations on matrix A.

$$\begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{(-1)\times R1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{\left(\frac{-1}{2}\right)\times R2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & (-1/2) \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{R4-R1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & (-1/2) \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

$$\xrightarrow{R3-R2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & (-1/2) \\ 0 & 0 & (3/2) \\ 0 & 0 & -2 \end{bmatrix} \xrightarrow{\left(\frac{2}{3}\right)\times R3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & (-1/2) \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{bmatrix} \xrightarrow{R4+2R3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & (-1/2) \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R1-R3}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & (-1/2)\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}\xrightarrow{R2+\left(\frac{1}{2}\right)R3}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}$$

The above matrix is in reduced row echelon form. From the above matrix, we can see that the only solution for a, b and c is $a = 0, b = 0$ and $c = 0$. Therefore, the given set of matrices are linearly independent.

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3. Question

A healthy juice consists of 30 units of protein, 11 units of carbohydrate, 53 units of fat, and 213 units of calcium. A juice maker makes two types of juice, Type A and Type B. Type A consists of banana, milk, and almond, whereas Type B consists of apple, milk, and almond. $Table\ M2W3G1$ shows the amount of protein, carbohydrate, fat, and calcium present in each banana, apple, and almond, and in 100 ml of milk.

Items	Protein	Carbohydrate	Fat	Calcium
Banana (1 piece)	2	3	1	5
Apple (1 piece)	1	2	1	6
Almond (1 piece)	6	1	15	1
Milk (100 ml)	4	1	3	100

Table: M2W3G1

Use the above information to choose the correct options.

- ☐ A healthy juice can be prepared with the right quantities of ingredients of Type A, and those quantities are unique.
- ☐ A healthy juice can be prepared with the right quantities of ingredients of Type B, and those quantities are unique.
- ☐ A healthy juice can not be prepared with the right quantities of ingredients of Type A.
- ☐ A healthy juice can not be prepared with the right quantities of ingredients of Type B.

3.1. Answers :

- A healthy juice can be prepared with the right quantities of ingredients of Type A, and those quantities are unique.
- A healthy juice can not be prepared with the right quantities of ingredients of Type B.

3.2. Solution

Let p, q, r and s denote the number of bananas, number of apples, number of almonds and number of 100

ml milk required to make a healthy juice, respectively.

<u>TYPE A</u>

To have a healthy juice of Type A, we need to satisfy the following equations : –

- **Protein:** $2p + 6r + 4s = 30$
- **Carbohydrate:** $3p + r + s = 11$
- **Fat:** $p + 15r + 3s = 53$
- **Calcium:** $5p + r + 100s = 213$

The above system of linear equations can be represented as $Ax = b$, where

$$A = \begin{bmatrix} 2 & 6 & 4 \\ 3 & 1 & 1 \\ 1 & 15 & 3 \\ 5 & 1 & 100 \end{bmatrix},\ x = \begin{bmatrix} p \\ r \\ s \end{bmatrix},\ b = \begin{bmatrix} 30 \\ 11 \\ 53 \\ 213 \end{bmatrix}$$

Augment A and b,

$$(A|b) = \left[\begin{array}{ccc|c} 2 & 6 & 4 & 30 \\ 3 & 1 & 1 & 11 \\ 1 & 15 & 3 & 53 \\ 5 & 1 & 100 & 213 \end{array}\right]$$

Apply elementary row operation on $(A|b)$

$$\left[\begin{array}{ccc|c} 2 & 6 & 4 & 30 \\ 3 & 1 & 1 & 11 \\ 1 & 15 & 3 & 53 \\ 5 & 1 & 100 & 213 \end{array}\right] \xrightarrow{R1/2} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 3 & 1 & 1 & 11 \\ 1 & 15 & 3 & 53 \\ 5 & 1 & 100 & 213 \end{array}\right] \xrightarrow{R2-3R1} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 0 & -8 & -5 & -34 \\ 1 & 15 & 3 & 53 \\ 5 & 1 & 100 & 213 \end{array}\right]$$

$$\xrightarrow{R3-R1} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 0 & -8 & -5 & -34 \\ 0 & 12 & 1 & 38 \\ 5 & 1 & 100 & 213 \end{array}\right] \xrightarrow{R4-5R1} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 0 & -8 & -5 & -34 \\ 0 & 12 & 1 & 38 \\ 0 & -14 & 90 & 138 \end{array}\right]$$

$$\xrightarrow{\frac{R2}{(-8)}} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 0 & 1 & (5/8) & (17/4) \\ 0 & 12 & 1 & 38 \\ 0 & -14 & 90 & 138 \end{array}\right] \xrightarrow{R3-12R2} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 0 & 1 & (5/8) & (17/4) \\ 0 & 0 & -(13/2) & -13 \\ 0 & -14 & 90 & 138 \end{array}\right]$$

$$\xrightarrow{R4+14R2} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 0 & 1 & (5/8) & (17/4) \\ 0 & 0 & -(13/2) & -13 \\ 0 & 0 & (395/4) & (395/2) \end{array}\right] \xrightarrow{\left(\frac{-2}{13}\right)\times R3} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 15 \\ 0 & 1 & (5/8) & (17/4) \\ 0 & 0 & 1 & 2 \\ 0 & 0 & (395/4) & (395/2) \end{array}\right]$$

$$\xrightarrow{R4-\left(\frac{395}{4}\right)R3}\left[\begin{array}{ccc|c}1&3&2&15\\0&1&(5/8)&(17/4)\\0&0&1&2\\0&0&0&0\end{array}\right]\xrightarrow{R1-2R3}\left[\begin{array}{ccc|c}1&3&0&11\\0&1&(5/8)&(17/4)\\0&0&1&2\\0&0&0&0\end{array}\right]\xrightarrow{R2-\left(\frac{5}{8}\right)R3}\left[\begin{array}{ccc|c}1&3&0&11\\0&1&0&3\\0&0&1&2\\0&0&0&0\end{array}\right]$$

$$\xrightarrow{R1-3R2}\left[\begin{array}{ccc|c}1&0&0&2\\0&1&0&3\\0&0&1&2\\0&0&0&0\end{array}\right]$$

The above matrix is in reduced row echelon form. We can see that the solution for $Ax = b$ is $:-$ $p = 2,\ r = 3,\ s = 2$. Hence, **option 1** is correct and **option 3** is incorrect.

<u>TYPE</u> <u>B</u>

To have a healthy juice of Type A, we need to satisfy the following equations $:-$

- **Protein:** $q + 6r + 4s = 30$
- **Carbohydrate:** $2q + r + s = 11$
- **Fat:** $q + 15r + 3s = 53$
- **Calcium:** $6q + r + 100s = 213$

The above system of linear equations can be represented as $Dx = f$, where

$$D = \begin{bmatrix}1&6&4\\2&1&1\\1&15&3\\6&1&100\end{bmatrix},\ x = \begin{bmatrix}q\\r\\s\end{bmatrix},\ f = \begin{bmatrix}30\\11\\53\\213\end{bmatrix}$$

Augment D and f,

$$(D|f) = \left[\begin{array}{ccc|c}1&6&4&30\\2&1&1&11\\1&15&3&53\\6&1&100&213\end{array}\right]$$

Apply elementary row operation on $(D|f)$

$$\left[\begin{array}{ccc|c}1&6&4&30\\2&1&1&11\\1&15&3&53\\6&1&100&213\end{array}\right]\xrightarrow{R2-2R1}\left[\begin{array}{ccc|c}1&6&4&30\\0&-11&-7&-49\\1&15&3&53\\6&1&100&213\end{array}\right]\xrightarrow{R3-R1}\left[\begin{array}{ccc|c}1&6&4&30\\0&-11&-7&-49\\0&9&-1&23\\6&1&100&213\end{array}\right]$$

$$\xrightarrow{R4-6R1}\left[\begin{array}{ccc|c}1&6&4&30\\0&-11&-7&-49\\0&9&-1&23\\0&-35&76&93\end{array}\right]\xrightarrow{R2+R3}\left[\begin{array}{ccc|c}1&6&4&30\\0&-2&-8&-26\\0&9&-1&23\\0&-35&76&93\end{array}\right]\xrightarrow[(-2)]{\frac{R2}{}}\left[\begin{array}{ccc|c}1&6&4&30\\0&1&4&13\\0&9&-1&23\\0&-35&76&93\end{array}\right]$$

$$\xrightarrow{R4+35R2}\left[\begin{array}{ccc|c}1&6&4&30\\0&1&4&13\\0&9&-1&23\\0&0&216&548\end{array}\right]\xrightarrow{R3-9R2}\left[\begin{array}{ccc|c}1&6&4&30\\0&1&4&13\\0&0&-37&-94\\0&0&216&548\end{array}\right]\xrightarrow{R4+6R3}\left[\begin{array}{ccc|c}1&6&4&30\\0&1&4&13\\0&0&-37&-94\\0&0&-6&-16\end{array}\right]$$

$$\xrightarrow{\frac{R3}{(-37)}}\left[\begin{array}{ccc|c}1&6&4&30\\0&1&4&13\\0&0&1&(94/37)\\0&0&-6&-16\end{array}\right]\xrightarrow{R46+R3}\left[\begin{array}{ccc|c}1&6&4&30\\0&1&4&13\\0&0&1&(94/37)\\0&0&0&6\times\left(\frac{94}{37}\right)-16\end{array}\right]$$

On converting the last column into a linear equation, we get $0q+0r+0s=6\times\left(\frac{94}{37}\right)-16$. For a linear equation of this form, we cannot have a solution. Hence, **option 2** is incorrect and **option 4** is correct.

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4. Question

Consider the set of vectors $S=\{(-1,1,5),(2,1,3),(2,1,2),(1,-1,7),(-1,3,-5)\}$ from $\mathbb{R}^3$ and choose the set of correct options.

- ☐ The singleton set $\{(-1,1,5)\}$ is linearly dependent.
- ☐ If $\alpha,\beta\in S$ and α,β are distinct then $\{\alpha,\beta\}$ is a linearly independent set of vectors.
- ☐ The set {(-1, 1, 5), (2, 1, 3),(-2, 2, 10)} is a linearly dependent set of vectors.
- ☐ The set S is a linearly independent set of vectors.
- ☐ The set $\{\alpha,\beta,\gamma\}$ is a linearly dependent set of vectors for any $\alpha,\beta,\gamma\in S$, where all the three are distinct vectors.
- ☐ The set $\{\alpha,\beta,\gamma,\delta\}$ is a linearly independent set of vectors for any $\alpha,\beta,\gamma,\delta\in S$, where all the four are distinct vectors.
- ☐ The system $AX=b$, where $A=\begin{bmatrix}2&1&2\\1&-1&7\\-1&1&5\\2&1&3\end{bmatrix}$ and $b=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$ has unique solution.
- ☐ The system $AX=b$, where $A=\begin{bmatrix}-1&3&-5\\2&1&3\end{bmatrix}$ and $b=\begin{bmatrix}0\\0\end{bmatrix}$ has infinitely many solutions.
- ☐ The system $AX=b$ where $A=\begin{bmatrix}2&1&2\\-1&3&-5\\-1&1&5\end{bmatrix}$ and $b=\begin{bmatrix}0\\0\\0\end{bmatrix}$ has infinitely many solutions.

4.1. Answers :

- If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.
- The set {(-1, 1, 5), (2, 1, 3),(-2, 2, 10)} is a linearly dependent set of vectors.
- The system $AX = b$, where $A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & -1 & 7 \\ -1 & 1 & 5 \\ 2 & 1 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ has unique solution.
- The system $AX = b$, where $A = \begin{bmatrix} -1 & 3 & -5 \\ 2 & 1 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ has infinitely many solutions.

4.2. Solution

- **Option 1** is incorrect. A set with only one vector is a linearly independent set. The only solution for $a \cdot (-1, 1, 5) = (0, 0, 0)$ is $a = 0$.

- **Option 2** is correct. Every pair of vectors in set S is linearly independent. For any two non-zero vectors v_1 *and* v_2 to be linearly dependent, the following equation must be satisfied : $av_1 + bv_2 = 0$, where $a, b \in \mathbb{R}$, $a \neq 0$ and $b \neq 0$.

  $av_1 + bv_2 = 0 \implies v_1 = \dfrac{-b}{a} v_2.$

  This shows that v_1 should be a scalar multiple of v_2, where $\dfrac{-b}{a}$ is a scalar.

  There is no vector in set S which is a scalar multiple of another vector from set S.

- **Option 3** is correct. Let us have a matrix M with the given vectors as its row vectors. If $det(M) = 0$, then the row vectors are not linearly independent.

$$M = \begin{bmatrix} -1 & 1 & 5 \\ 2 & 1 & 3 \\ -2 & 2 & 10 \end{bmatrix}$$

$$\begin{aligned} det(M) &= -1(10-6) - 2(10-10) - 2(3-5) \\ det(M) &= -1(4) - 2(-2) \\ det(M) &= 0 \end{aligned}$$

  Since $det(M) = 0$, the given set of vectors are not linearly independent.

- **Option 4** is incorrect. Let us have matrix A with all the vectors in its columns.

$$A = \begin{bmatrix} -1 & 2 & 2 & 1 & -1 \\ 1 & 1 & 1 & -1 & 3 \\ 5 & 3 & 2 & 7 & -5 \end{bmatrix}$$

  If $Ax = 0$ has a unique solution, then the column vectors of matrix A are independent.
  We can see that there are more variables than equations for $Ax = 0$. There are 3 rows (3 equations) and 5 columns (5 variables or 5 unknowns). As a result, we have infinite solutions for $Ax = 0$. In

other words $x = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ is not the only solution. This is true for homogeneous systems of linear equations only. For non-homogeneous systems of linear equations, this is not always true. Therefore, set S is a linearly dependent set. In a 3 dimensional space, there can atmost be three linearly independent vectors. The given set has 5 vectors, therefore, the set is not linearly independent.

- **Option 5** is incorrect. Let us take $\alpha = (-1, 1, 5), \beta = (2, 1, 3)\ and\ \gamma = (2, 1, 2)$. Let matrix B have $\alpha, \beta\ and\ \gamma$ in its columns. If $det(B) \neq 0$, then these three vectors are linearly independent

$$B = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 1 \\ 5 & 3 & 2 \end{bmatrix}$$

$$\begin{aligned} det(B) &= -1(2-3) - 2(2-5) + 2(3-5) \\ det(B) &= 1 + 6 - 4 \\ det(B) &= 3 \end{aligned}$$

Since $det(B) \neq 0$, the column vectors of B are independent.

- **Option 6** is incorrect. Let us take $\alpha = (-1, 1, 5), \beta = (2, 1, 3), \gamma = (2, 1, 2)$ and $\delta = (1, -1, 7)$. Let matrix C have $\alpha, \beta, \gamma\ and\ \delta$ in its columns.

$$C = \begin{bmatrix} -1 & 2 & 2 & 1 \\ 1 & 1 & 1 & -1 \\ 5 & 3 & 2 & 7 \end{bmatrix}$$

For $Cx = 0$, we have more columns than rows. As a result, we have infinite solutions for $Cx = 0$. In other words $x = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ is not the only solution. This is true for homogeneous systems of linear equations only. For non-homogeneous systems of linear equations, this is not always true. Therefore, the column vectors of matrix C are not linearly independent.

- **Option 7** is correct. Let us convert matrix A to reduced row echelon form.

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & -1 & 7 \\ -1 & 1 & 5 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow{R1/2} \begin{bmatrix} 1 & 1/2 & 1 \\ 1 & -1 & 7 \\ -1 & 1 & 5 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow{R2-R1} \begin{bmatrix} 1 & 1/2 & 1 \\ 0 & -3/2 & 6 \\ -1 & 1 & 5 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow{R3+R1} \begin{bmatrix} 1 & 1/2 & 1 \\ 0 & -3/2 & 6 \\ 0 & 3/2 & 6 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow{R3+R2}\begin{bmatrix}1 & 1/2 & 1\\0 & -3/2 & 6\\0 & 0 & 12\\2 & 1 & 3\end{bmatrix}\xrightarrow{R3/12}\begin{bmatrix}1 & 1/2 & 1\\0 & -3/2 & 6\\0 & 0 & 1\\2 & 1 & 3\end{bmatrix}\xrightarrow{R2-6R3}\begin{bmatrix}1 & 1/2 & 1\\0 & -3/2 & 0\\0 & 0 & 1\\2 & 1 & 3\end{bmatrix}\xrightarrow{\left(\frac{-2}{3}\right)\times R2}\begin{bmatrix}1 & 1/2 & 1\\0 & 1 & 0\\0 & 0 & 1\\2 & 1 & 3\end{bmatrix}$$

$$\xrightarrow{R1-\frac{R2}{2}}\begin{bmatrix}1 & 0 & 1\\0 & 1 & 0\\0 & 0 & 1\\2 & 1 & 3\end{bmatrix}\xrightarrow{R1-R3}\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\\2 & 1 & 3\end{bmatrix}\xrightarrow{R4-2R1}\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\\0 & 1 & 3\end{bmatrix}\xrightarrow{R4-R2}\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\\0 & 0 & 3\end{bmatrix}\xrightarrow{R4-3R3}\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix}$$

The above matrix is in reduced row echelon form. There are no independent variables. When we augment this matrix with b, we don't get any row of the form $\left[\begin{array}{ccc|c}0 & 0 & 0 & c\end{array}\right]$, where $c \neq 0$. This ensures consistency of $AX = b$. When a system of linear equations is consistent and its reduced row echelon form has no independent variables, we say that it has a unique solution.

- **Option 8** is correct. $AX = b$ is a homogeneous system of linear equations. A homogeneous system of linear equations has infinite solutions if it has more unknowns than the number of equations.
- **Option 9** is incorrect. If A is invertible, i.e., if $det(A) \neq 0$, then $AX = b$ has a unique solution.

$$A = \begin{bmatrix}2 & 1 & 2\\-1 & 3 & -5\\-1 & 1 & 5\end{bmatrix}$$

$$\begin{aligned}det(A) &= 2(15-(-5)) - 1(-5-(5)) + 2(-1-(-3))\\det(A) &= 2(20) - 1(-10) + 2(2)\\det(A) &= 54\end{aligned}$$

Since $det(A) \neq 0$, $AX = b$ has a unique solution and not infinite solutions.

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5. Question

Consider the set of three vectors $S = \{(1, 2, -1), (3, 1, 0), (-1, 2, c)\}$ in $\mathbb{R}^3$ with usual addition and scalar multiplication. For which value of c will the above set S be linearly dependent?

5.1. Answer : –1.4

5.2. Solution

Let matrix A have all the given vectors in its column.

$$A = \begin{bmatrix}1 & 3 & -1\\2 & 1 & 2\\-1 & 0 & c\end{bmatrix}$$

The column vectors would be linearly dependent if the $det(A) = 0$.

$$\begin{aligned}det(A) &= -1(6-(-1)) + c(1-6)\\det(A) &= -7 + 5c\end{aligned}$$

Let us equate $det(A)$ with 0

$$\begin{aligned} -7 - 5c &= 0 \\ c &= \frac{-7}{5} \\ c &= -1.4 \end{aligned}$$

If $c = -1.4$, then set S will be linearly dependent.

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6. Question

Consider the set of three vectors $S = \{(9, 2, 8), (7, 5, 9), (13, -4, c)\}$ in $\mathbb{R}^3$ with usual addition and scalar multiplication. If S is a linearly independent set, then the value of c can not be equal to

6.1. Answer : 6

6.2. Solution

Let matrix A have all the given vectors in its column.

$$A = \begin{bmatrix} 9 & 7 & 13 \\ 2 & 5 & -4 \\ 8 & 9 & c \end{bmatrix}$$

The column vectors would be linearly independent if the $det(A) \neq 0$.

$$\begin{aligned} det(A) &= 8((-28) - (65)) - 9(-36 - 26) + c(45 - 14) \\ det(A) &= 8(-93) - 9(-62) + c(31) \\ det(A) &= 31c - 186 \end{aligned}$$

Let us solve the inequality $det(A) \neq 0$

$$\begin{aligned} 31c - 186 &\neq 0 \\ 31c &\neq 186 \\ c &\neq 6 \end{aligned}$$

If $c \neq 6$, then set S will be linearly independent.

Comprehension $(7, 8, 9, 10)$

In genetics, a classic example of dominance is the inheritance of shape of seeds in peas. Peas may be round (associated with genotype R) or wrinkled (associated with genotype r). In this case, three combinations of genotypes are possible: RR, rr, and Rr. The RR individuals have round peas and the rr individuals have wrinkled peas. In Rr individuals the R genotype masks the presence of the r genotype, so these individuals also have round peas.

Consider the crossing of RR with RR. This always gives the genotype RR, therefore the probabilities of an offspring to be RR, Rr, and rr respectively are equal to 1, 0, and 0. Next, consider the crossing of Rr with

RR. The offspring will have equal chances to be of genotype RR and genotype Rr, therefore the probabilities of RR, Rr, and rr respectively are 1/2, 1/2, and 0. Finally, consider the crossing of rr with RR. This always results in genotype Rr. Therefore, the probabilities of genotypes RR, Rr, and rr are 0, 1, and 0, respectively.

The following table summarizes these facts :

Parents' genotypes	Genotypes of offspring		
	RR	Rr	rr
RR-RR	1	0	0
RR-Rr	$\frac{1}{2}$	$\frac{1}{2}$	0
RR-rr	0	1	0

Table: M2W3G2

The matrix representing this observation is given by $P = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix}$, and the initial distribution vector (1×3 *matrix*) is denoted by $X_0 = \left(X_0^1, X_0^2, X_0^3\right)$, where X_0^1 denotes the distribution of RR, X_0^2 denotes the distribution of Rr, and X_0^3 denotes the distribution of rr. For any positive integer n, the distribution vector after n generations ($i.e.,$ at $t = n$) is denoted by X_n and is given by the equation $X_{n-1}P = X_n$. Using the above information, answer the following questions.

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7. Question

Suppose, in an experiment, 100 pairs of parents with genotype combinations RR-RR, 100 pairs of parents with genotype combinations RR-Rr, and 200 pairs of parents with genotype combination RR-rr are taken to observe the genotypes of their offspring. Suppose from crossing of each pair of parents a single offspring is produced. Find the set of correct options from the following.

- ☐ There will be at least 200 offspring with wrinkled peas.
- ☐ There will be no offspring with wrinkled peas.
- ☐ There will be no offspring with round peas.
- ☐ All the offspring will have round peas.
- ☐ All the offspring will have wrinkled peas.
- ☐ There will be at least 100 offspring with combination of genotypes RR.
- ☐ There will be at least 200 offspring with combination of genotypes Rr.

7.1. Answers :

- There will be no offspring with wrinkled peas.
- All the offspring will have round peas.
- There will be at least 100 offspring with combination of genotypes RR.
- There will be at least 200 offspring with combination of genotypes Rr.

7.2. Solution

- Offsprings with genotype rr alone have wrinkled peas. However, the probability of having an offspring with genotype rr is zero for all the three types of crossings. There will be no offspring with genotype rr, therefore no offspring with wrinkled seed. This would also mean that all the offsprings of RR-RR crossing, RR-Rr crossing and RR-rr crossing will result in a round seed. Hence, **option 1** is incorrect, **option 2** is correct, **option 3** is incorrect, **option 4** is correct and **option 5** is incorrect.
- All the offsprings of RR-RR crossing will have RR genotype. Half of RR-Rr crossing will result in offsprings with RR genotype and half of RR-Rr crossing will result in offsprings with Rr genotype. All the offsprings of RR-rr crossing will have Rr genotype.
- From the above observation, we can see that all the hundred offsprings of RR-RR crossing will have RR genotype. 50 of RR-Rr crossing will have RR genotype. 50 of RR-Rr crossing will have Rr genotype. All the 200 offsprings of RR-rr crossing will have Rr genotype. Approximately, 150 offsprings will have RR genotype and 250 offsprings will have Rr genotype. Hence, **option 6** and **option 7** are correct.

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8. Question

Which of the following options are correct?

- ☐ The rows of the matrix P form a linearly independent set.
- ☐ $\{(\frac{1}{2}, 0, 0), (0, \frac{1}{2}, 0)\}$ is a linearly independent set in X.
- ☐ The columns of the matrix P form a linearly dependent set.
- ☐ $det(P)$ is a non-zero number.

8.1. Answers :

- $\{(\frac{1}{2}, 0, 0), (0, \frac{1}{2}, 0)\}$ is a linearly independent set in X.
- The columns of the matrix P form a linearly dependent set.

8.2. Solution

- The last column of matrix P has no non-zero entry. $det(P) = 0$. The rows of matrix P are not linearly independent. Hence, **option 1** is incorrect, **option 3** is correct and **option 4** is incorrect.

- $a \cdot \left(\frac{1}{s}, 0, 0\right) + b \cdot \left(0, \frac{1}{2}, 0\right) = (0, 0, 0) \implies \left(\frac{a}{2}, \frac{b}{2}, 0\right) = (0, 0, 0)$. We can see that $a = 0$ and $b = 0$ is the only solution. The given set is a linearly independent set. Hence, **option 2** is correct.

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9. Question

Suppose $X_0 = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. Find out the correct set of correct options.

- ☐ X_0 and X_1 are linearly dependent.
- ☐ X_0 and X_1 are linearly independent.
- ☐ The set $\{X_0, X_1, X_2\}$ is a linearly dependent set.
- ☐ The set $\{X_0, X_1, X_2\}$ is a linearly independent set.

9.1. Answers :

- X_0 and X_1 are linearly independent.
- The set $\{X_0, X_1, X_2\}$ is a linearly independent set.

9.2. Solution

- $X_{n-1}P = X_n \implies X_0P = X_1$.

$$X_1 = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

- Let us solve $a \cdot X_0 + b \cdot X_1 = 0$.

$$\begin{aligned} a\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) + b\left(\frac{1}{2}, \frac{1}{2}, 0\right) &= (0, 0, 0) \\ \left(\frac{a}{3} + \frac{b}{2}, \frac{a}{3} + \frac{b}{2}, \frac{a}{3}\right) &= (0, 0, 0) \\ \left(\frac{2a + 3b}{6}, \frac{2a + 3b}{6}, \frac{a}{3}\right) &= (0, 0, 0) \qquad (1) \end{aligned}$$

From (1) we get,

$$\frac{2a + 3b}{6} = 0 \implies 2a + 3b = 0 \qquad (2)$$

$$\frac{a}{3} = 0 \implies a = 0 \qquad (3)$$

Substitute (3) in (2),

$$2(0) + 3b = 0 \implies 3b = 0 \implies b = 0$$

$a = 0$ and $b = 0$ is the only solution for $a \cdot X_0 + b \cdot X_1 = 0$. Therefore X_0 and X_1 are linearly independent. Hence, **option 1** is incorrect and **option 2** is correct.

- $X_{n-1}P = X_n \implies X_1P = X_2$.

$$X_2 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \end{bmatrix}$$

- Let us solve $p \cdot X_0 + q \cdot X_1 + r \cdot X_2 = 0$.

$$\begin{aligned} p\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) + q\left(\frac{1}{2}, \frac{1}{2}, 0\right) + r\left(\frac{3}{4}, \frac{1}{4}, 0\right) &= (0, 0, 0) \\ \left(\frac{p}{3} + \frac{q}{2} + \frac{3r}{4}, \frac{p}{3} + \frac{q}{2} + \frac{r}{4}, \frac{p}{3}\right) &= (0, 0, 0) \\ \left(\frac{4p + 6q + 9r}{12}, \frac{4p + 6q + 3r}{12}, \frac{p}{3}\right) &= (0, 0, 0) \quad (4) \end{aligned}$$

From (4) we get,

$$\frac{4p + 6q + 9r}{12} = 0 \implies p + 6q + 9r = 0 \quad (5)$$

$$\frac{4p + 6q + 3r}{12} = 0 \implies 4p + 6q + 3r = 0 \quad (6)$$

$$\frac{p}{3} = 0 \implies p = 0 \quad (7)$$

Substitute $p = 0$ in (5),

$$\begin{aligned} 6q + 9r &= 0 \\ 6q &= -9r \quad (8) \end{aligned}$$

Substitute $p = 0$ in (6),

$$\begin{aligned} 6q + 3r &= 0 \\ 6q &= -3r \quad (9) \end{aligned}$$

From (8) and (9) we get.

$$\begin{aligned} -9r &= -3r \\ -r &= 0 \\ r &= 0 \end{aligned}$$

On substituting $r = 0$ in (8), we get $q = 0$

$p = 0$, $q = 0$ and $r = 0$ is the only solution for $p \cdot X_0 + q \cdot X_1 + r \cdot X_2 = 0$. Therefore X_0, X_1 and X_2 are linearly independent. Hence, **option 3** is incorrect and **option 4** is correct.

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10. Question

Suppose $X_0 = (a, b, 1 - a - b)$, where $0 \leq a, b, a + b \leq 1$. If the set $\{X_0, X_2\}$ is linearly dependent, then what is the value of b?

10.1. Answer : 0

10.2. Solution

$X_{n-1}P = X_n$. From $X_0P = X_1$ and $X_1P = X_2$, we get $X_0 \times P \times P = X_2 \implies X_0P^2 = X_2$.

$$X_2 = \begin{bmatrix} a & b & 1-a-b \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} a & b & 1-a-b \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$X_2 = \left[a + \frac{3b}{4} + \frac{1-a-b}{2} \,\middle|\, \frac{b}{4} + \frac{1-a-b}{2} \,\middle|\, 0 \right]$$

$$X_2 = \left[\frac{4a + 3b + 2 - 2a - 2b}{4} \,\middle|\, \frac{b + 2 - 2a - 2b}{4} \,\middle|\, 0 \right]$$

$$X_2 = \left[\frac{2a + b + 2}{4} \,\middle|\, \frac{2 - 2a - b}{4} \,\middle|\, 0 \right] \qquad (1)$$

It is given that X_0 and X_2 are linearly dependent.

For $\mathbf{pX_0 + qX_2 = 0}$, $p = 0$ and $q = 0$ is not the only solution. Therefore, we get $X_2 = \frac{-p}{q} X_0$, for $q \neq 0$. Let $k = \frac{-p}{q}$. We get $X_2 = kX_0$.

From (1) and $X_2 = kX_0$ we get,

$$\left(\frac{2a + b + 2}{4}, \frac{2 - 2a - b}{4}, 0 \right) = k(a, b, 1 - a - b)$$

$$\left(\frac{2a+b+2}{4}, \frac{2-2a-b}{4}, 0\right) = (ka, kb, 1-a-b) \quad (2)$$

From (2) we get:

$$\begin{aligned} \frac{2a+b+2}{4} &= ka \\ 2a+b+2 &= 4ka \quad (3) \end{aligned}$$

$$\begin{aligned} \frac{2-2a-b}{4} &= kb \\ 2-2a-b &= 4kb \quad (4) \end{aligned}$$

$$\begin{aligned} 0 &= k(1-a-b) \\ 0 &= k-ka-kb \\ -k &= -ka-kb \\ 1 &= a+b \\ 1-a &= b \quad (5) \end{aligned}$$

Substitute (5) in (3),

$$\begin{aligned} 2a+(1-a)+2 &= 4ka \\ a+1+2 &= 4ka \\ 3 &= a(4k-1) \quad (6) \end{aligned}$$

Substitute (5) in (4),

$$\begin{aligned} 2-2a-(1-a) &= 4k(1-a) \\ 2-2a-1+a &= 4k-4ka \\ 1-a &= 4k-4ka \\ 4ka-a &= 4k-1 \\ a(4k-1) &= 4k-1 \quad (7) \end{aligned}$$

Substitute (6) in (7),

$$\begin{aligned} 3 &= 4k-1 \\ 4 &= 4k \\ k &= 1 \end{aligned}$$

Substitute $k = 1$ in (6),

$$\begin{aligned} 3 &= a(4(1)-1) \\ 3 &= a(3) \\ a &= 1 \end{aligned}$$

Substitute $a = 1$ in (5),

$$
\begin{aligned}
1 - (1) &= b \\
b &= 0
\end{aligned}
$$